

Between-Session Depression Monitoring for NUS Students

A Data-Driven Mental Health Intervention System

DSA4262 Group 2

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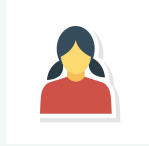
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User Persona - Mei Ling

User Demographic



- 21-year-old Year 3 undergraduate at NUS
- Diagnosed with **moderate depression**
- Currently enrolled in UHC counselling programme
- Attends counselling sessions every 7-14 days

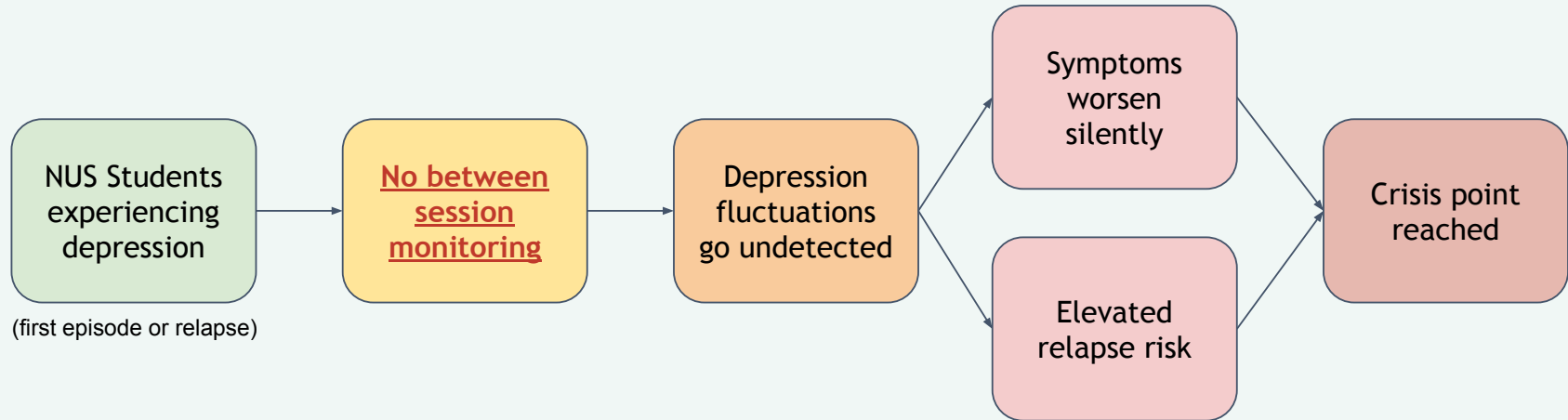
- No monitoring between sessions
- Lack of real-time communication channel with counsellors
- Uncertainty about when to seek help
- Counselling sessions are **too infrequent** to capture evolving symptoms
- Experiences **gaps in care** during critical periods of deterioration



*“Some days, I spiral really fast... I don't know what to do. **The next session feels so far away. The system only sees me when I'm functional enough to show up**, not when I actually need help the most.”*

Problem Statement

“NUS students experiencing depression, whether in their first episode or managing a relapse, struggle to receive timely and proportionate mental health support within the current UHC counselling framework. This is due to the absence of any between-session monitoring mechanism, meaning that fluctuations in mental state go undetected and unaddressed between appointments. This results in unmonitored symptom deterioration, elevated risk of relapse and an increased likelihood of reaching a crisis point before the students' next scheduled session.”



Jobs to Be Done

	Early Warning Detection	Receiving Adequate Care at the Right Time
What Is The JTBD?	• Detect worsening depressive symptoms between counselling sessions • Identify early warning signs before escalation • Receive timely intervention without waiting for next appointment	• Receive timely and appropriate support during difficult periods • Access care that matches symptom severity • Avoid waiting for fixed scheduled appointments
Design Rationale	• 7-14 day gaps create monitoring blind spots • No way to signal deterioration to care team • Early symptoms go unnoticed till next session • Students face uncertainty navigating distress alone	• Support is not responsive to real-time needs • Students with higher severity receive insufficient care • Depressive episodes are dynamic and unpredictable

Data-Driven Motivation

IHME Global Burden Of Disease

About The Dataset

- Nationally representative data on depression prevalence in Singapore
- Focus on 15-24 age group (university-aged population)

Link To Problem Statement

- Identifies youth (15 to 24) as a high-prevalence demographic
- Reinforces need for targeted intervention in university settings

Reddit & Twitter Severity Levels

About The Dataset

- Collection of Reddit & Twitter posts labelled by depression severity
- Captures linguistic patterns, sentiment and emotional tone

Link To Problem Statement

- Shows that depressive symptoms are often expressed informally and indirectly
- Highlights gap in systems that rely on self-reporting or clinical visits only

CBT Dataset

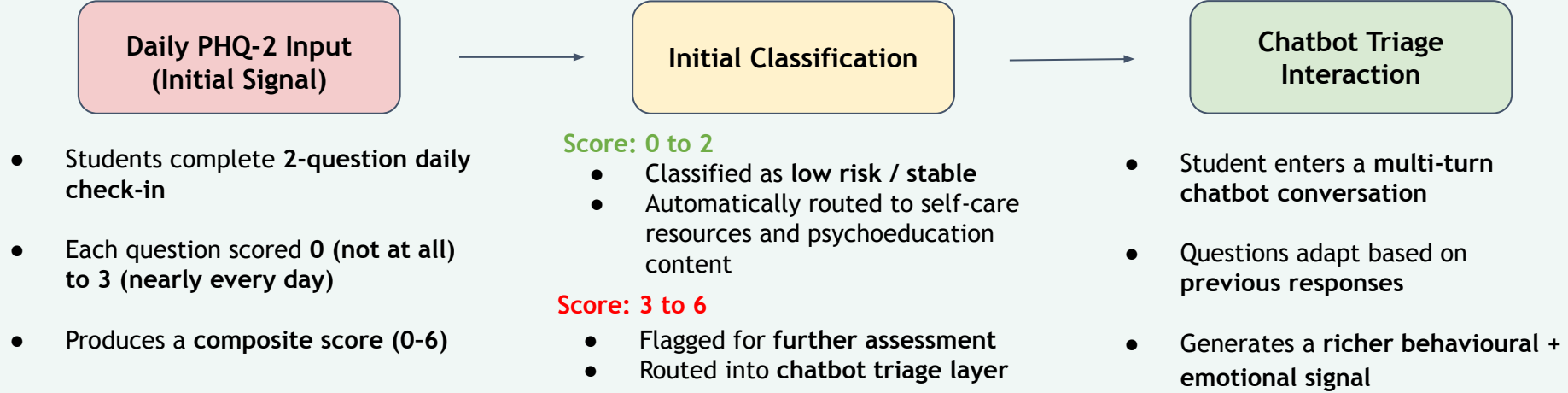
About The Dataset

- Clinical dataset from CBT treatment for depression
- Includes PHQ scores across session frequency and waiting times

Link To Problem Statement

- Reveals gaps between sessions and their impact on outcomes
- Demonstrates how delays in intervention correlate with worsening symptoms

PHQ-2 Daily Check-In



For The Chatbot Triage Interaction Model

NLP-Based Severity Classification:

- Chat responses are concatenated and analysed by our **NLP classifier**
- Outputs a **severity score (0 to 3 scale)**

Decision & Intervention Routing:

- NLP severity score determines final pathway**
- Enables more accurate escalation decision and appropriate intervention.

Signal-to-Decision Table

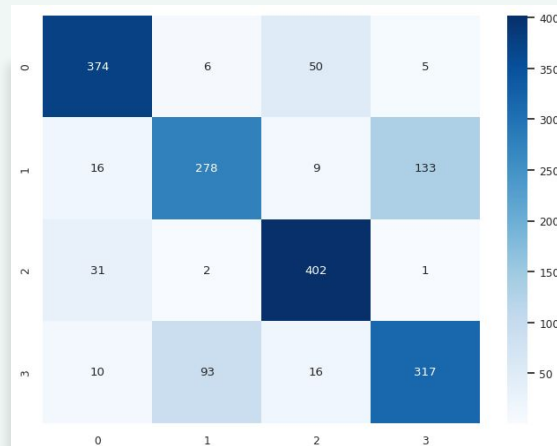
Tier	PHQ-2 Score	Chatbot NLP Score	Outcome
1 (Well)	0 to 2	—	• Chatbot triage NOT activated • Provide self-care resources (psychoeducation, mindfulness)
2 (Grey Zone -> Stable)	3 to 6	0	• Chatbot triage activated • Provide self-care resources and provide emergency hotlines
3 (Grey Zone -> Mild)	3 to 6	1	• Chatbot triage activated • Provide option to book appointment with counsellor ◦ Students without an appointment can book a new one ◦ Student with an upcoming appointment can push it forward • Provide self-care resources and provide emergency hotlines
4 (Grey Zone -> Moderate)	3 to 6	2	• Chatbot triage activated • Provide option to speak to counsellor immediately via live chat • Provide option to book appointment with counsellor (same as tier 3) • Provide self-care resources and provide emergency hotlines
5 (Grey Zone -> Critical)	3 to 6	3	• Chatbot triage activated • Provide option to speak to counsellor immediately via live chat • Provide option to book appointment with counsellor (same as tier 3) • Provide self-care resources and provide emergency hotlines

NLP Model for Grey Zone Triage



Fine-Tuned BERT (Selected Model)

- PEFT (LoRA) on bert-base-cased
- Predict depression severity from a given textual input (integer score from 0-3)
- 41,859 labelled Reddit/Twitter samples
- 10 epochs, learning rate 1×10^{-4}



Confusion Matrix (Classification)

Confusion Matrix Insights (Per-Class Recall)

Class 2 (moderate) → ~92% recall (402/436) — strongest performer

Class 0 (stable) → ~86% recall (374/435) — reliable for low-risk classification

Class 3 (severe) → ~73% recall (317/436) — some confusion with Class 1

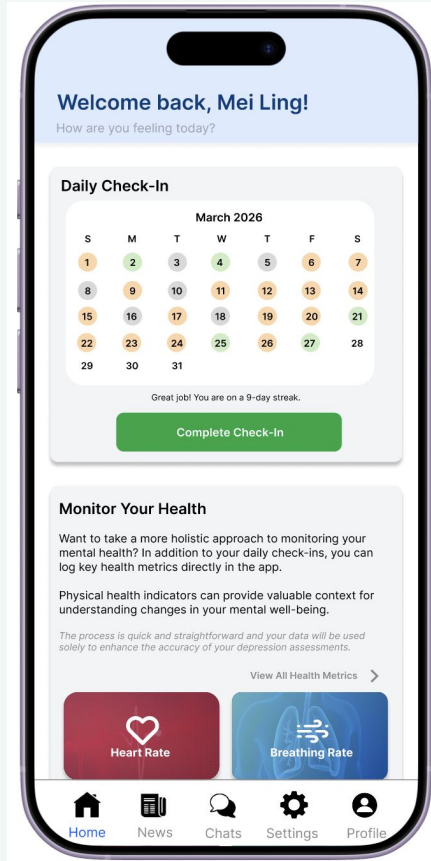
Class 1 (mild) → ~64% recall (278/436) — most confused with Class 3 (bidirectional)

Design choice: When uncertain, system defaults to flagging for counsellor review (erring on safety)

Model Comparison

	Accuracy	F1
Classification	0.787	0.784
Regression	0.564	0.561

Overview Of Product Interface



- Designed in Figma as a mobile application
- Clean, structured interface with a calm visual language appropriate for a mental health context
- Pilot-phase deployment targeting NUS UHC-enrolled students with depression
- Features:
 - a. Daily PHQ-2 check-in
 - b. Chatbot Triage
 - c. Appointment Booking
 - d. News & Resources

Human-in-the-Loop Design

Targeted support at every tier, depending on severity

Tier 1-2 Stable	Tier 3 Mild	Tier 4 Moderate	Tier 5 Severe
Student receives: • Self-care resources • Emergency hotlines Counsellor role: NIL	Student receives: • Option to book appointment • Self-care resources • Emergency hotlines Counsellor role: Student-initiated contact only	Student receives: • Live chat option with human counsellor • Option to book appointment • Self-care resources • Emergency hotlines Counsellor role: Dashboard alerts for counsellor review and response	Student receives: • Live chat option with human counsellor • Option to book appointment • Self-care resources • Emergency hotlines Counsellor role: Real-time escalation for immediate support
Autonomous	Autonomous	Human Review	Immediate Escalation

Counsellor can confirm, override, or escalate any system alert at any tier.

Scenario 1: Low Risk

Example: Tier 3 (Mild)

1

Daily PHQ-2 check-in

Mei Ling scores 4 for PHQ-2 → Chatbot triage activated

2

Dynamic questioning by Chatbot

Mei Ling responds in her own words via free-text response

3

NLP model classifies: Tier 3 (Mild)

- Based on Mei Ling's text responses, the NLP model deems that she falls under *Tier 3 (Mild)*
- Not in crisis, but not entirely okay either

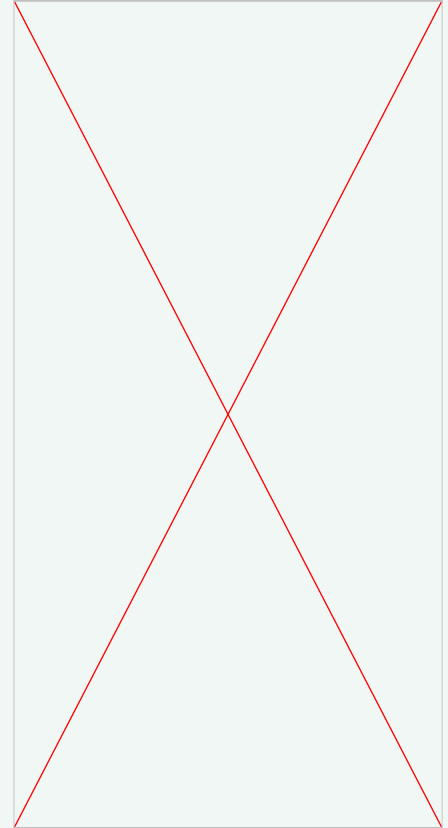
4

Targeted support delivered

- Option to book an earlier appointment slot
- Curated self-care resources
- Emergency hotline always available

Mei Ling gets targeted help between-sessions

Early detection allows Mei Ling to seek help earlier, before the 8-day wait becomes a crisis



Scenario 2: High Risk

Example: Tier 5 (Severe)

1

Daily PHQ-2 check-in

Mei Ling scores 6 for PHQ-2 → Chatbot triage activated

2

Dynamic questioning by Chatbot

Mei Ling responds in her own words via free-text response

3

NLP model classifies: Tier 5 (Severe)

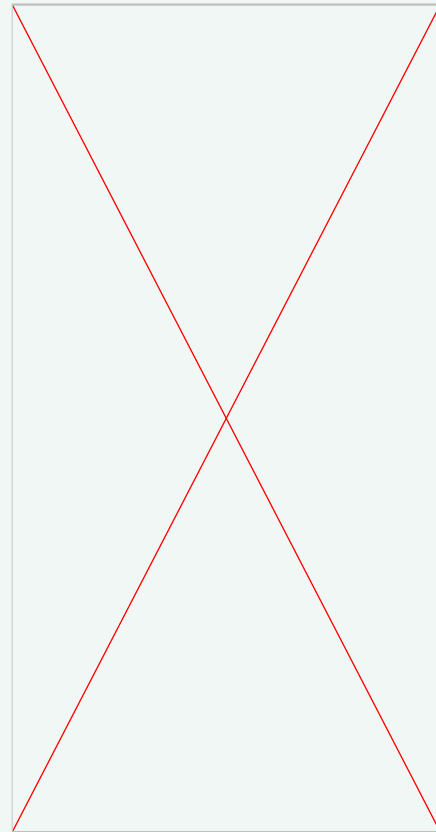
- Based on Mei Ling's text responses, the NLP model deems that she falls under *Tier 5 (Severe)*
- Mei Ling is in acute distress and needs help now

4

Targeted support delivered

- Option to speak to a human counsellor via live chat
- Option to book an earlier appointment slot
- Curated self-care resources
- Emergency hotline always available

Mei Ling gets targeted help between-sessions
Connects with her counsellor right when she needs help the most, instead of having to wait 8 days



Success Metrics

Reduction in Response Delay for High-Severity and Deteriorating Students

How It Works?

- Measures time elapsed between a Tier 4/5 escalation flag and the delivery of counsellor intervention or priority appointment
- Pre-rollout delay approximated using CBT baseline dataset, where deteriorating students had to wait the remaining interval of the fixed 7-14 day schedule
- Tier 1 (stable) response delays monitored to verify that improvement is due to reallocation, not additional resources

Comparison

- Post-rollout Tier 4/5 response delay vs pre-rollout approximated delay of 7-14 days
- Tier 1 delays tracked to confirm counsellor capacity is being reallocated rather than expanded

Success Criteria

- Tier 4/5 average response delay reduces significantly, ideally to same-day or within 1-2 days
- Tier 1 response times remain broadly stable, confirming effective triage reallocation

PHQ Improvement Relative to Baseline, Controlling for Natural Recovery

How it works

- Students grouped by initial PHQ-9 severity band: mild (5-9), moderate (10-14), severe (15-19)
- Average PHQ delta (first minus last session score) calculated per band, controlling for session count and programme duration
- Matched baseline cohort drawn from CBT dataset: same severity band, same session count, no check-in system access

Comparison

- Post-rollout PHQ delta vs matched baseline cohort within each severity band
- Holds session count constant to isolate the effect of intervention timing rather than volume of therapy received

Success Criteria

- Moderate and severe band students show meaningfully greater PHQ improvement than matched baseline peers
- Mild band showing smaller or negligible difference is expected and consistent with design intent

Intended Use and Scope of Model and Application



Who is this system designed for?

- **Pilot phase:** exclusively for NUS students enrolled in UHC counselling for depression and their assigned UHC counsellors
- **Phase 2:** expand to all Singapore university students (NTU, SMU, SIT and others)
- **Phase 3:** broaden to all Singapore youths aged 15 to 24, including those outside universities
- Continuous monitoring of model performance and clinical outcomes required at every phase



What this system is explicitly not

- Not a replacement for professional therapy, counsellors retain full authority over all clinical decisions
- Not a standalone diagnostic tool where outputs inform rather than replace formal clinical assessment
- Designed to reduce pressure on UHC by decreasing dependency on physical appointments for stable cases



How is access granted & managed?

- Restricted to students already enrolled in UHC's counselling programme (no independent self-registration)
- Access granted on an opt-in basis following UHC's standard intake process
- Counsellor accounts provisioned directly by UHC, tied to their active caseload
- System operates strictly as an extension of an existing supervised clinical relationship

Handling High-Risk & Outlier Cases

Anomaly 1: Repeated High NLP Severity Scores

- A single high PHQ-2 score (5 to 6) doesn't auto-escalate; sustained high NLP severity scores (2 to 3) over multiple days trigger human review
- The system flags these cases to a counsellor's dashboard, prompting them to review the student's history and reach out directly

Anomaly 2: Non-Response to Check-ins

- Prolonged silence is treated as potential "silent deterioration": the app first sends a gentle, supportive nudge (not a demanding one)
- If the student remains unresponsive, the case escalates to a counsellor who reaches out personally to check on the student's well-being

Reducing Misclassification of Worsening Cases

- The NLP chatbot asks a fixed number of free-text questions. If it can't reach a confident conclusion, it defaults to flagging the student for counsellor review
- The system is deliberately biased toward false positives (unnecessary check-ins) over false negatives (missing a student in crisis) since the consequences of the latter are far more serious

Data Privacy & Consent

Data Collection and Retention

- Two main data types are collected: daily PHQ-2 check-in scores and free-text chatbot responses. Voluntary health indicators (sleep, heart rate, breathing rate etc.) are also stored if provided
- Data is retained only while the student is actively enrolled. Once discharged, all data is permanently and securely deleted

Access Controls

- Only the student's assigned UHC counsellor can access their data: no other staff, counsellors or third parties have access
- A strict minimum-access principle is enforced, with data siloed at the individual counsellor level

Consent

- Consent is obtained in person during onboarding before any data is collected, with the counsellor explaining what is collected, why and who sees it
- Consent is fully voluntary and can be withdrawn at any time, immediately halting data collection and triggering secure disposal of existing data

Compliance Policies

- The system operates in full compliance with Singapore's PDPA and NUS's internal data governance policies
- The highest data protection standards are applied throughout, ensuring the handling of student health data is lawful, transparent and proportionate

Limitations Of Current Prototype

PHQ-2 as Primary Screening Tool

- The PHQ-2's two questions are too limited for comprehensive screening and key factors like family history and environmental stressors go uncaptured
- This weakens triage reliability, risking false positives and false negatives

Unoptimized Model Explainability

- Error analysis provides initial insights, but more advanced techniques are needed to better trace predictions
- Current insights are useful but insufficient to fully confirm the model's decision-making robustness process

No Real-World Pilot

- The system has not been tested with real users, meaning actual performance, trust, engagement and response honesty remain uncertain
- Scaling beyond NUS would introduce further variability in language, cultural attitudes and potential misuse that cannot be anticipated without pilot data

Implementation Barriers



Training Data and Platform Mismatch (Technical)

- The model is trained on Reddit and Twitter data (informal, anonymous and emotionally unfiltered) but students using the app are likely to write more formally, knowing a counsellor may review their responses
- This mismatch reduces the model's ability to detect distress in restrained or measured language, raising the risk of false negatives, particularly for students whose communication style already diverges from the training data



Counsellor Capacity Constraints (Organisational)

- UHC counsellors support students with different mental health conditions, not just depression
- If there is a high number of cases that are flagged as high-risk, this could potentially lead to counsellor fatigue and delayed responses if not managed carefully



Cultural and Linguistic Bias (Cultural)

- The training data is largely Western-centric, meaning the model may not recognise Singlish or culturally specific phrases (e.g. "sian", "damn shag") that NUS students commonly use to express distress
- Asian cultural norms also tend toward indirect or understated expressions of difficulty which the model may miss entirely or misinterpret, reducing overall classifier reliability

Ethical Concerns

1

Over-Reliance on Automated Triage

- Students may treat the NLP classifier as a substitute for professional help, delaying or avoiding counsellor support
- Especially dangerous if false negatives give at-risk students a false sense of security
- In a diverse environment like NUS, over-reliance could further discourage vulnerable students from engaging with formal support, undermining the system's intended purpose

2

Psychological Impact of Classification Labels

- Being flagged as "Critical" without context or immediate human follow-up may cause unnecessary anxiety, self-stigma or feelings of hopelessness
- Misclassification risks either alarming students needlessly or failing to validate real distress. Outputs should always be framed supportively and paired with clear guidance and counsellor access

Future Work

Longitudinal PHQ Tracking

Establish personalised baselines per student to better detect sustained depressive trends versus temporary stress, improving early identification accuracy

Electronic Health Record Integration

Link PHQ and NLP data to UHC's clinical records, giving counsellors richer context (example: treatment history) for more informed and efficient decision-making

Improve Model Explainability

Apply techniques like LIME, SHAP or BERT attention maps to surface the specific language cues driving each classification, increasing transparency for counsellors

Implementing Real-World Pilot Study

Deploy with a small diverse cohort at UHC to validate performance, refine alert thresholds and surface biases before broader rollout

Broader Mental Health Scope

Extend screening beyond depression to cover anxiety and stress comorbidities, enabling more holistic and tailored interventions

Richer Input features

Supplement PHQ scores and free-text with biometric data (heart rate, blood pressure, oxygen saturation) for a more comprehensive assessment of mental health severity

Thank You

*Bridging the gap between sessions,
before students reach crisis point.*

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EDA 1: Depression Prevalence by Age Group

Percentage of People with Depressive Disorders in Singapore By Age Group



Type Of Visualization Used

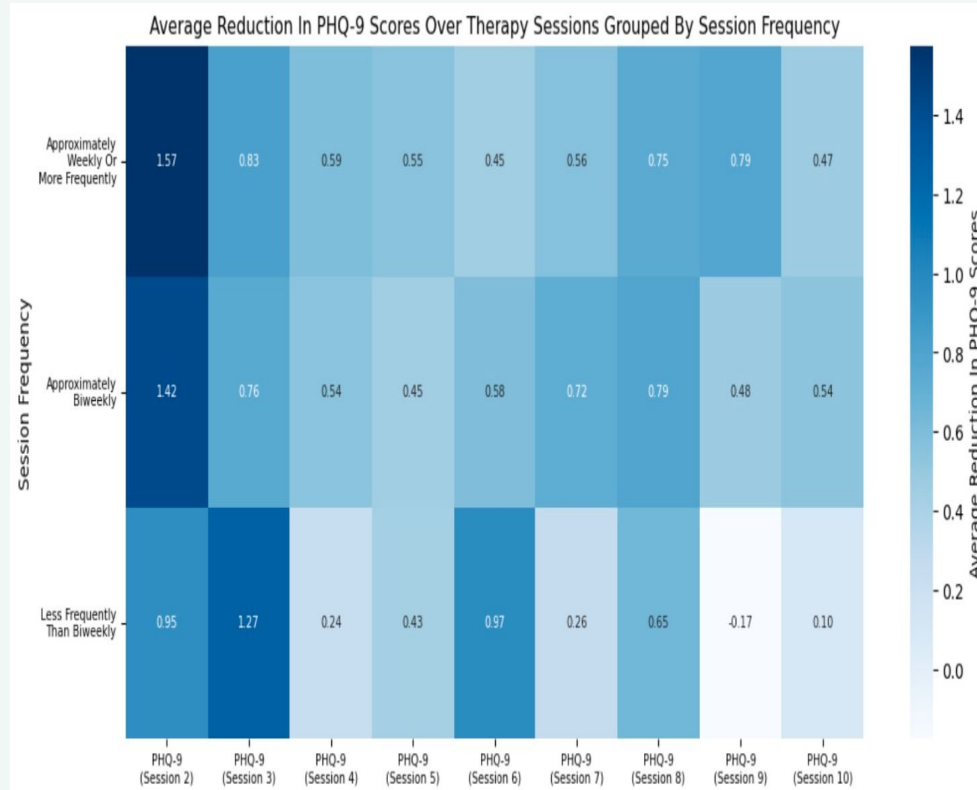
- Heatmap of depression prevalence rates in Singapore
- Displays **age groups** (y-axis) against **year** (x-axis)
- Colour intensity represents **severity of prevalence**

Key Observations

- **15-24 age group** consistently shows the highest prevalence across all years
- Peak observed at **2.54%** in 2021, with only a slight decline to **2.48%** in 2023
- Clear **spike between 2020-2021**, indicating sensitivity to external shocks (example: COVID-19)
- Minimal recovery post-pandemic suggests **persistent, not temporary mental health impact**
- Other age groups show **lower and more stable prevalence patterns**

- **NUS students (15 to 24)** fall directly within the **highest-risk demographic**
- University life stage compounds risk due to **academic pressure, social transition and career uncertainty**

EDA 2: Session Frequency vs PHQ-9 Reduction



Type Of Visualization Used

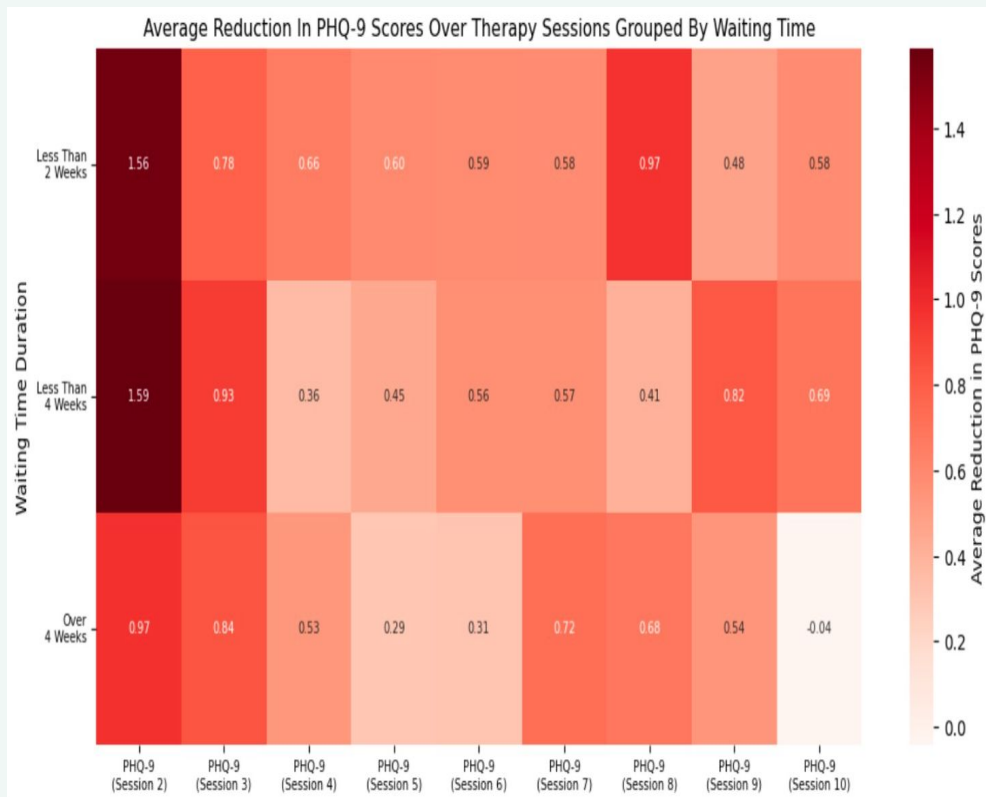
- **Heatmap** showing average reduction in PHQ-9 scores across therapy sessions
- Displays **Session frequency** vs **Session number**
- Colour intensity represents **magnitude of symptom improvement**

Key Observations

- **Weekly sessions** show the most **consistent and sustained reductions** in PHQ-9 scores
- **Biweekly sessions** demonstrate moderate improvement but with **less consistency over time**
- **Less frequent sessions** show **highly irregular outcomes**, with fluctuations across sessions
- Largest improvements occur **early (Session 2 to 3)** across all groups, highlighting importance of **early engagement**

- Students under the **bi-weekly UHC model** face **inconsistent treatment outcomes**
- Extended gaps increase risk of **unmonitored deterioration and relapse**

EDA 3: Waiting Time vs PHQ-9 Reduction



Type Of Visualization Used

- **Heatmap** showing average reduction in **PHQ-9** scores across therapy sessions
- Displays **Waiting Time** vs **Session Number**
- Colour intensity represents **magnitude of symptom improvement**

Key Observations

- Patients who started therapy **within 2 weeks** show the **strongest and most consistent reductions**
 - **2 to 4 week wait group** demonstrates **moderate but less stable improvements**
 - **>4 week wait group** shows the **weakest overall reductions across sessions**
 - Early sessions (Session 2 to 3) show **initial improvement across all groups**, but divergence appears over time
- **Delays in access to care are clinically consequential**
 - Longer waiting times increase risk of **symptom worsening before intervention begins**
 - Early intervention leads to **stronger and more stable recovery trajectories**